distinct choices or selections are $A, B, C; A, B, D; A, C, D$ and B, C, D. We may thus conclude that there are 4 ways of selecting 3 people out of 4. Each choice or selection is referred to as a combination of 4 different objects taken 3 at a time.

Suppose two persons are to be selected from four persons. The possible choices are: $A, B : A, C : A, D : B, C : B, D : C, D$. Thus the number of combinations of 4 different objects taken 2 at a time is 6. The number of combinations of n different objects taken r at a time is represented by ${}^nC_r$. From the above we may conclude that ${}^4C_3 = 4$ and ${}^4C_2 = 6$. Now, ${}^4C_3$ is the number of combinations of 4 objects taken 3 at a time. Note that in each combination, the three objects may be arranged in 3! ways. Thus the total number of permutations of 4 objects taken 3 at a time is ${}^4C_3 \times 3!$. This is also equal to ${}^4P_3$. Hence ${}^4P_3 = {}^4C_3 \times 3!$.

In general, this leads to an important relationship between permutations and combinations as,

$$ {}^nP_r = {}^nC_r \times r!. $$

Normally for any reader there may be a confusion between permutation and combination. The following table with an example may be helpful in clearing the confusion.

S.No	Description	Permutation	Combination
1	What is a	Number of Arrangement or Listing of objects	Number of Selections or Grouping of objects
2	Where to use	If the ordering of objects matters	If the ordering of objects does not matter
3	Representation	${}^nP_r$	${}^nC_r$
		Examples	
4	In a game of cricket	The number of batting line up of 11 players out of the 15 players	The number of teams consisting of 11 players out of 15 players
5	In a process of prize distribution	The number of ways of distributing 3 distinct prizes	The number of ways of distributing 3 identical prizes
6	In a committee formation	The number of ways of choosing a President and a Vice-President for a committee of 13 members	The number of ways of forming a committee of 2 persons from 13 members
7	In a process of choosing objects	The number of ways of choosing 3 out of 15 distinct objects one after another	The number of ways of choosing 3 out of 15 distinct objects simutaneously

Theorem 4.5: The number of combinations of n distinct objects taken r at a time is given by

$$ {}^nC_r = \frac{n!}{r!(n-r)!}, 0 \leq r \leq n. $$